THESIS PROPOSAL — UPDATED

A Neuro-Symbolic Decision Layer over Time-Series Foundation Model (TSFM) Embeddings

Primary Evaluation Domain: Automotive CAN Bus Anomaly Detection

Mohamad
moea00781@gmail.com

August 2026 — Updated following Phase 1 & Phase 2 completion

✥ Changes from the June 2026 original are marked in green throughout.

General Summary

This document presents the updated thesis proposal for a research project combining a frozen pre-trained time-series foundation model (MOMENT) with a JEPA-style predictive concept layer to produce accurate and inspectable anomaly detection for automotive CAN bus networks. The project bridges three fields — time-series representation learning, self-supervised predictive architectures, and automotive cybersecurity — that have remained largely separate in the literature.

> ***Core research question:*** *Can a JEPA-style predictive concept layer over frozen MOMENT embeddings support accurate and interpretable CAN bus anomaly detection?*

The architecture has four sequential components. First, a frozen MOMENT encoder transforms raw CAN signal windows into patch-level embeddings without any in-domain training. Second, a multi-resolution JEPA prediction module learns what normal CAN traffic looks like by predicting masked latent embeddings at multiple time scales. Third, a soft concept/prototype bottleneck clusters the JEPA latent space into K human-inspectable regimes of normal behaviour. Fourth, an explicit rule-based decision layer raises anomaly alerts when prediction error is high, the active concept is rare, and this state persists across multiple windows.

This architecture replaces the original ANFIS/DFL/LTN formulation, which was impractical for a thesis effort and would have produced fuzzy rules over a thousand-dimensional embedding space that are not meaningfully interpretable. The JEPA approach is scientifically stronger, better aligned with current self-supervised representation learning, and produces genuinely human-readable decision rules operating over learned latent concepts.

Evaluation uses the SynCAN public benchmark with a six-step ablation ladder that isolates the contribution of each architectural component. Cross-domain validation on battery health and industrial fault datasets confirms that the frozen MOMENT backbone generalises beyond the automotive domain.

1. Introduction and Problem Statement

1.1 The CAN Bus and the Anomaly Detection Challenge

The Controller Area Network (CAN) bus is the communication backbone of modern vehicles. Every electronic control unit — engine management, braking, steering, infotainment — broadcasts messages on this shared network.

❖ Corrected: SynCAN has 10 CAN IDs and 20 signals total (not 281 channels). Signal counts per ID: 2,3,2,1,2,2,2,1,1,4. Messages arrive asynchronously at 15 ms / 30 ms / 45 ms intervals and must be resampled to a uniform 15 ms zero-order-hold grid before windowing.

❖ Corrected: SynCAN has 5 attack types (not 3): plateau, suppress, flooding, continuous, and playback. The dataset uses a file-based split (train_1–4, test_normal, and 5 attack test files) — not a scenario-based split.

Anomaly injection on the CAN bus — whether from a cyberattack or a sensor fault — can produce safety-critical consequences at the network level. Five canonical attack types are studied in SynCAN:

Attack	Description	Difficulty
Plateau	Signal frozen at abnormal constant value	Medium
Suppress	Legitimate messages silenced or dropped	Hard
Flooding	Bus saturated with high-frequency injected messages	Medium
Continuous	Continuous signal manipulation	Medium
Playback	Replayed legitimate messages injected	Medium

1.2 The Interpretability Requirement

Current state-of-the-art ML anomaly detectors — LSTM autoencoders, variational autoencoders, Transformer-based models — produce scalar anomaly scores with no traceable decision path. A safety engineer examining an alert cannot answer: which signal triggered this? what pattern was unusual? how long did it persist? This opacity creates a practical barrier for deploying ML-based intrusion detection in production vehicles.

> ***Thesis objective:*** *match the accuracy of black-box detectors while simultaneously producing traceable, inspectable decisions — using a JEPA predictive concept layer that makes the anomaly reasoning process explicit at the level of learned latent behaviour regimes.*

The interpretability goal is grounded in Rudin (2019): the thesis builds an inherently interpretable model, not a post-hoc explainer. SHAP and Grad-CAM are explicitly excluded from the design.

2. Literature Review

The thesis draws on five bodies of literature. Papers below were read and annotated in full; their relevance to specific architectural components is noted.

2.1 Time-Series Foundation Models (TSFMs)

MOMENT — Goswami et al. (2024)

MOMENT is the primary backbone. It is a T5-based transformer pre-trained on approximately 27 billion time-series tokens. The model processes input series as sequences of fixed-length patches (patch_len = 8 timesteps), producing a 1,024-dimensional embedding per patch. Available via HuggingFace as AutonLab/MOMENT-1-large. Its frozen weights serve as the feature extraction backbone.

❖ Verified: MOMENT-1-large has exactly 341,240,320 parameters, all frozen. Forward pass: [batch, 20, 512] → per-channel patch reps [B, 20, 64, 1024] → mean-pool over 20 channels → [B, 64, 1024].

PatchTST — Nie et al. (2023)

PatchTST introduced patch-based tokenisation for time-series transformers, motivating the patch_len = 8 design in MOMENT. Also the primary fallback backbone if MOMENT embedding quality is insufficient.

2.2 Predictive Self-Supervised Architectures (JEPA)

JEPA — LeCun (2022)

The Joint Embedding Predictive Architecture (JEPA) trains a predictor to forecast masked representations in latent embedding space — not in signal space. In this thesis, the JEPA module is trained on normal CAN traffic: it learns to predict what a masked MOMENT embedding should look like. High prediction error at inference is a direct anomaly signal.

I-JEPA — Assran et al. (2023)

I-JEPA demonstrated that JEPA-style masked latent prediction produces representations superior to contrastive and pixel-reconstruction baselines. Key implementation principle used in this thesis: contiguous block masking (not random patch masking), which forces genuine temporal prediction rather than interpolation.

Concept Bottleneck Models — Koh et al. (2020)

CBM provides the structural analogy for the prototype bottleneck layer. The two-stage CBM training protocol maps directly onto JEPA pre-training followed by rule extraction.

Background: ANFIS, DFL, LTN

The original proposal used ANFIS (Jang 1993), Differentiable Fuzzy Logic (van Krieken 2022), and Logic Tensor Networks (Badreddine 2022). These were replaced following supervisor feedback: fuzzy rules over a 1,024-dimensional embedding space are not meaningfully interpretable. These papers are retained as background references only — not implemented.

2.3 CAN Bus Anomaly Detection

CANet — Hanselmann et al. (2020) [Primary Baseline]

State-of-the-art CAN bus intrusion detection baseline. Trains a separate LSTM autoencoder per CAN signal channel on normal traffic only. Weakness: per-channel modelling with no cross-channel correlation, no inspectable decision logic.

❖ Corrected evaluation protocol: CANet paper reports per-message stream AUC (plateau 0.982, suppress 0.743). These numbers are NOT reproducible at window level (W=512, stride=512) and are reference only. Validated window-level CANet baselines: plateau 0.854 · suppress 0.790 · flooding 0.840 · continuous 0.680 · playback 0.654.

Bohne et al. (2023) [Closest Prior Work]

Knowledge-graph-guided FCN on automotive oscillograms. Explicitly states as future work that their method cannot handle multivariate, multi-ID time series — the exact setting addressed in this thesis.

2.4 General Time-Series Anomaly Detection Baselines

Paper	Type	Role
Anomaly Transformer (Xu 2022)	Transformer + assoc. discrepancy	Evaluated on SynCAN + Phase 6
USAD (Audibert 2020)	Dual-AE adversarial	Evaluated on SynCAN + Phase 6
OmniAnomaly (Su 2019)	Stochastic RNN + norm. flows	Evaluated on Phase 6 datasets

3. Research Gaps and Contributions

3.1 Identified Gaps

Gap 1 — No TSFM applied to CAN bus anomaly detection

Existing CAN detectors train from scratch. MOMENT's potential for transfer has not been explored.

Gap 2 — No JEPA-style predictive layer bridging TSFM embeddings to inspectable decisions

No prior work applies JEPA-style latent prediction to CAN signals or uses prediction error as a human-readable anomaly signal.

Gap 3 — No concept-level interpretability for multivariate CAN anomaly detection

Existing detectors produce scalar scores. Bohne (2023) uses hand-coded rules on univariate signals only.

3.2 Thesis Contributions

>#	Contribution	Key novelty
1	First application of a frozen TSFM to CAN bus anomaly detection	Directly addresses Bohne et al. 2023 future-work statement.
2	JEPA predictive concept layer over TSFM embeddings	First JEPA-style masked latent prediction on CAN signals. Trained on normal traff
3	Soft concept/prototype bottleneck for latent regime discovery	K learned prototypes cluster JEPA latent space into human-inspectable driving reg
4	Explicit rule-based decision layer over concept activations	Final decisions by human-readable IF-THEN rules — not opaque neural activation

4. Proposed Architecture

Pipeline: CAN Bus Signals → MOMENT Encoder (frozen) → Multi-resolution JEPA Module → Concept/Prototype Bottleneck → Explicit Decision Rules → Anomaly Score + Rule Trace

4.1 Input Representation

Input is a multivariate CAN bus window comprising all signal channels for a fixed time window.

✥ Corrected: W = 512 timesteps (not variable). N_channels = 20 (not 281). The asynchronous CAN bus (15/30/45 ms message periods) must first be resampled to a uniform 15 ms zero-order-hold grid, then z-score normalised per signal using train_1–3 statistics. Window shape fed to MOMENT: [batch, 20, 512].

4.2 Layer 1 — Frozen MOMENT Encoder

MOMENT processes each window and produces patch-level embeddings. All weights are frozen throughout training — no gradients pass through the encoder at any point.

✥ Verified: Input [batch, 20, 512] → MOMENT internal per-channel patches [B, 20, 64, 1024] → mean-pool over 20 channels → output [B, 64, 1024]. Total frozen params: 341,240,320. Trainable params: 0. Confirmed on RTX 5050 (sm_120, cu128 build).

4.3 Layer 2 — Multi-Resolution JEPA Prediction Module

The JEPA module learns normal CAN traffic patterns by predicting masked patch embeddings. High prediction error at inference = anomaly signal. Multi-resolution processing is essential because different attacks manifest at different time scales:

Branch	Mask size	Target attack	Rationale	Params
Short	6 patches	Flooding	Abrupt bus saturation — visible in small window	~3.7M
Medium	24 patches	Suppress	Signal absence detectable over medium window	~3.7M
Long	38 patches	Plateau	Signal freeze requires large masked block	~3.7M

✥ Implemented and smoke-tested: 3-branch MultiResolutionJEPA, total ~11M params. Each branch: 4-layer context encoder + 2-layer predictor, d_model=256, n_heads=4. Masking: contiguous block masking only (no random scatter). Targets are .detach()ed — no gradients leak into MOMENT. compute_prediction_error() returns {'short': float, 'medium': float, 'long': float}.

Training loss: MSE between predicted and actual patch embeddings in MOMENT representation space. Never in signal space — following the I-JEPA principle.

4.4 Layer 3 — Soft Concept / Prototype Bottleneck

The JEPA latent space is clustered into K learned prototype vectors representing recurring regimes of normal CAN behaviour (e.g., 'highway cruise', 'urban stop-go', 'engine idle'). Each window is assigned a soft membership score over all K prototypes. Concept rarity = $1 - \max(s_k)$ — rises toward 1 when the window matches no known prototype.

Concept validation is mandatory: for each prototype, extract the 10 most-assigned normal windows, plot raw CAN traces, assign plain-language labels, and report Jensen-Shannon divergence across 5 independent runs (target: JSD < 0.05).

4.5 Layer 4 — Explicit Rule-Based Decision Layer

Final anomaly decisions are made by explicit human-readable rules:

IF prediction_error IS HIGH [≥ 95th percentile on normal val set]
AND concept_rarity IS HIGH [≥ 95th percentile on normal val set]
AND this state PERSISTS for ≥ P windows
THEN anomaly_score = w1·norm(pred_error) + w2·concept_rarity + w3·persistence_flag

Thresholds calibrated on train_4 (held-out normal validation set) at the 95th percentile. Weights w1, w2, w3 set by logistic regression or equal weighting. No backpropagation in Stage 3 — purely calibration.

4.6 Trainable Parameters Summary

Component	Status	Parameters
MOMENT Encoder	FROZEN — never updated	341,240,320 (not trained)
JEPA Prediction Module	Trainable — Stage 1	~11M (3 branches × ~3.7M)
Concept Prototype Bottleneck	Trainable — Stage 2	K × 256 (K = 8, 16, or 32)
Decision Rule Weights	Calibrated — Stage 3	< 10 scalar parameters

5. Training Protocol

5.1 Data Split and Training Assumption

Training uses normal traffic only — no labelled anomaly samples are used at any point during backpropagation. This is the realistic deployment assumption.

❖ Corrected split: train_1–3 → training (normal only, 12,126 windows); train_4 → validation (normal only, 4,042 windows, used for threshold calibration); test_normal + 5 attack test files → evaluation (586 windows each). No scenario-based split exists in SynCAN — it is file-based.

5.2 Stage 1 — JEPA Pre-training on Normal Traffic

The three JEPA branches are trained on normal CAN traffic by masking contiguous blocks of MOMENT patch embeddings and training each predictor to reconstruct them from context. MOMENT is frozen throughout. Loss: MSE in embedding space.

Optimiser: Adam, lr=1e-3, weight_decay=1e-4. Batch size tuned to VRAM. Training data: train_1–3 normal windows only.

5.3 Stage 2 — Concept Bottleneck Learning

K prototype vectors are initialised via k-means++ on JEPA latent representations of all normal training windows. The bottleneck is fine-tuned with JEPA. Loss: L_reconstruction + λ · L_diversity (diversity loss prevents prototype collapse).

5.4 Stage 3 — Decision Rule Calibration (no backprop)

Prediction error, concept rarity, and persistence count computed over train_4 (normal val). Thresholds set at 95th percentile of each signal. Anomaly threshold on final score: 95th percentile of anomaly_score on train_4.

6. Evaluation Plan

6.1 Primary Benchmark: SynCAN — Window-Level Protocol

❖ Evaluation protocol clarified: all models evaluated at window level (W=512, stride=512 at test time). Per-message stream AUC from the CANet paper (plateau 0.982, suppress 0.743) is NOT reproducible at window level and is documented as reference only. All reported numbers in this thesis use window-level AUC.

6.2 Six-Step Ablation Ladder

>Step	System	Purpose
1	MOMENT-only anomaly score	Lower bound — L2 distance from normal centroid in embedding space
2	MOMENT + simple neural head	Supervised upper bound (if attack labels available)
3	MOMENT + one-class detector (OC-SVM / Deep SVDD)	Unsupervised baseline, no temporal structure
4	MOMENT + JEPA prediction error only	Does JEPA alone suffice without the concept layer?
5	MOMENT + JEPA + concept bottleneck	Does concept rarity add signal over raw prediction error?
6	Full: MOMENT + JEPA + concept + persistence	Does persistence improve suppress attack detection?

✥ Step 1 complete. MOMENT-only window-level AUC: plateau 0.521 · suppress 0.600 · flooding 0.540 · continuous 0.463 · playback 0.432. t-SNE confirms heavy class overlap in raw embeddings — validates low Step 1 AUC and motivates the JEPA module. Saved to experiments/ablation_results.csv.

6.3 Window-Level Performance Baselines

✥ Updated: 5 attack types evaluated (not 3). All targets are window-level AUC, not per-message stream AUC.

Attack	CANet window-level AUC	Step 1 (MOMENT-only)	Target
Plateau	0.854	0.521	≥ 0.900
Suppress	0.790	0.600	> 0.790
Flooding	0.840	0.540	> 0.840
Continuous	0.680	0.463	> 0.680
Playback	0.654	0.432	> 0.654

6.4 Comparison Baselines

Baseline	Type	Key weakness
CANet (Hanselmann 2020)	Per-ID LSTM-AE	No cross-channel correlation; not interpretable
CANTransfer (Hanselmann 2021)	ConvLSTM + domain adapt	Not interpretable; vehicle-specific retraining required
Anomaly Transformer (Xu 2022)	Transformer + assoc. discrepancy	General-purpose; no CAN-specific structure
USAD (Audibert 2020)	Dual-AE adversarial	Reconstruction-only; no concept layer
OmniAnomaly (Su 2019)	Stochastic RNN + flows	Probabilistic but not inspectable

6.5 Concept Validation Metrics (mandatory)

- Concept stability: Jensen-Shannon divergence between concept assignment distributions across 5 independent runs (target: JSD < 0.05).
- Concept correspondence: for each prototype, plot the 10 most-assigned normal windows and manually label the driving regime.

- Attack discriminativeness: report concept assignment shift between normal and attack windows per attack type.
- Rule understandability: ask two non-ML domain experts to interpret the printed IF-THEN rules.

6.6 Evaluation Metrics

- AUC-ROC — primary ranking metric, threshold-independent.
- Average Precision (AP) — area under the precision-recall curve.
- F1 Score — at optimally-selected threshold on validation set.
- Detection Delay — windows from attack start to first alert.
- Concept Stability — JSD across 5 runs.

6.7 Cross-Domain Validation

Domain	Dataset	Task	Target AUC
Battery Health	Severson et al. 2019 (data.matr.io/1)	Detect early degradation	≥ 0.70
Industrial Faults	CWRU Bearing Dataset	Bearing faults vs. normal	≥ 0.70 on ≥ 3/4 faults

Only the JEPA module and concept bottleneck are retrained per domain. MOMENT stays frozen — enabling zero-cost transfer of the backbone.

7. Development Timeline

❖ Status as of August 2026: Phase 1 COMPLETE · Phase 2 COMPLETE · Phase 3 in progress (Sections A & B done, Sections C–F remaining).

Phase	Weeks	Goal	Status
1 — Environment & Baseline	1–2	Data pipeline + CANet reproduction	<font color="#186B3A">■ COMPLETE</fon
2 — MOMENT Integration	3–4	Frozen encoder + embedding validation	<font color="#186B3A">■ COMPLETE</fon
3 — JEPA Module	5–7	Multi-resolution JEPA pre-training	<font color="#C06A00">■ IN PROGRESS</
4 — Concept Bottleneck	7–8	Prototype learning + concept validation	■ Pending
5 — Full System + Ablation	9–10	6-step ablation on SynCAN	■ Pending
6 — Cross-Domain + Writing	11–12	Battery + CWRU evaluation; thesis draft	■ Pending

Gate	Week	Deliverable	Status
Checkpoint 1	2	t-SNE of MOMENT embeddings + Step 1 AUC results	<font color="#186B3A">■ READY
Checkpoint 2	5	JEPA training curves + prediction error distributions + Step 4 AUC	■ Pending
Checkpoint 3	10	Full ablation table + cross-domain AUC + concept validation report	■ Pending

8. Risk Assessment and Mitigation

Risk	Likelihood	Severity	Mitigation
MOMENT embeddings do not separate normal from attack	Low	High	t-SNE at Checkpoint 1 confirmed — heavy overlap expected at th
JEPA fails to learn stable latent prediction on CAN	Medium	High	Use I-JEPA contiguous block masking; fall back to future-frame pr
Concept prototypes unstable across seeds	Medium	Medium	k-means++ initialisation; try K=8,16,32; report JSD stability metric
Step 4 ≈ Step 6 (rules don't help)	Medium	Low	Honest null result; contribution shifts to JEPA + concept layer (Ste
Cross-domain AUC < 0.70	Low	Low	Conservative targets; partial failure still validates primary CAN res

9. Technical Stack and Resources

9.1 Software

Package	Version	Role
Python	3.14.0	Primary language
PyTorch	2.11.0+cu128	All neural modules — cu128 required for RTX 5050 (sm_120)
momentfm	0.1.4 (--no-deps)	MOMENT encoder — must install without deps
transformers	5.12.1	MOMENT internals
huggingface-hub	1.21.0	Model download + caching
scikit-learn	1.9.0	k-means++, AUC metrics, OC-SVM baseline
numpy / pandas	2.4.4 / 3.0.2	Data loading, preprocessing, results aggregation
matplotlib / seaborn	3.10.9 / 0.13.2	t-SNE, prediction error curves, concept heatmaps
scipy	1.18.0	Signal processing utilities

9.2 Datasets

Dataset	Source	Phase
SynCAN	github.com/etas/SynCAN	Phase 1–5 (primary)
Battery Health	data.matr.io/1	Phase 6 — Domain 2A
CWRU Bearing	engineering.case.edu/bearingdatacenter	Phase 6 — Domain 2B
ROAD (optional)	0xsam.com/road	Phase 5 — secondary automotive validation

10. References

[1] Assran, M., et al. (2023). Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture (I-JEPA). CVPR 2023.

[2] Audibert, J., et al. (2020). USAD: UnSupervised Anomaly Detection on Multivariate Time Series. KDD 2020.

[3] Badreddine, S., et al. (2022). Logic Tensor Networks. Artificial Intelligence, 303, 103649. [background]

[4] Bohne, T., et al. (2023). Knowledge Graph-guided Anomaly Detection for Automotive Signals. IEEE Trans. ITS.

[5] Dong, X., et al. (2023). SimMTM: A Simple Pre-Training Framework for Masked Time-Series Modeling. NeurIPS 2023.

[6] Goswami, M., et al. (2024). MOMENT: A Family of Open Time-series Foundation Models. ICML 2024.

[7] Guo, S., et al. (2022). A Transformer-based Approach for Bearing Fault Diagnosis (DCTLN). IEEE Trans. Instrumentation and Measurement.

[8] Hanselmann, M., et al. (2020). CANet: An Unsupervised Intrusion Detection System for High Dimensional CAN Bus Traffic. IEEE Access.

[9] Hanselmann, M., et al. (2021). CANTransfer: Transfer Learning based Intrusion Detection on a Controller Area Network. SAC 2021.

[10] Jang, J. S. R. (1993). ANFIS: Adaptive-Network-based Fuzzy Inference System. IEEE Trans. SMC. [background]

[11] Koh, P. W., et al. (2020). Concept Bottleneck Models. ICML 2020.

[12] LeCun, Y. (2022). A Path Towards Autonomous Machine Intelligence. (JEPA framework white paper.)

[13] Longari, S., et al. (2021). CANnolo: Anomaly Detection System based on LSTM Autoencoders for CAN. IEEE Trans. NSM.

[14] Nie, Y., et al. (2023). A Time Series is Worth 64 Words: Long-term Forecasting with Transformers (PatchTST). ICLR 2023.

[15] Rudin, C. (2019). Stop Explaining Black Box Machine Learning Models for High Stakes Decisions. Nature Machine Intelligence, 1, 206–215.

[16] Severson, K. A., et al. (2019). Data-driven prediction of battery cycle life before capacity degradation. Nature Energy, 4(5), 383–391.

[17] Su, Y., et al. (2019). Robust Anomaly Detection for Multivariate Time Series through Stochastic Recurrent Neural Network. KDD 2019.

[18] van Krieken, E., et al. (2022). Analyzing Differentiable Fuzzy Logic Operators. Artificial Intelligence, 302, 103602. [background]

[19] Verma, A., et al. (2022). ROAD: The Real ORNL Automotive Dynamometer CAN Intrusion Detection Dataset. IEEE Trans. ITS.

[20] Xu, J., et al. (2022). Anomaly Transformer: Time Series Anomaly Detection with Association Discrepancy. ICLR 2022.

[21] Yue, Z., et al. (2022). TS2Vec: Towards Universal Representation of Time Series. AAAI 2022.